MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,
SCHOTTKY BARRIER, FAST RECOVERY,
TYPE 1N6391
JAN, JANTX, AND JANTXV

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon, fast recovery, Schottky barrier semiconductor diode, intended for use as a power rectifier in rectifier circuits, or as a flyback diode in power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1. DO-4.

1.3 Maximum ratings.

Type	V_{RRM} and V_{RWM} 1/	V_{RSM}	V_R 1/	I_{FM} 2/ T_C = 125°C	I_O 3/ T_C = 125°C	I_{FSM}	T_{op} T_{STG}
	V (pk)	V (pk)	V dc	A dc	A dc	A (pk)	°C
1N6391	45	54	45	25	22.5	600	-55 to +175

1/ Full rated V_{RRM} and V_{RWM} is applicable over the range of T_C = -55°C to +139°C for I_{FM} = 0. Full rated V_R is applicable over the temperature range of T_C = -55°C to +139°C. When V_R = 45 V dc and T_C = +139°C, then T_J = 175°C.

2/ Average current with a 50% duty cycle square wave including reverse amplitude equal to the magnitude of full rated V_{RWM}. Derate linearly at 0.64 A dc/°C for T_C>125°C.

3/ Average current with an applied sine wave peak value equal to the magnitude of full rated V_{RWM}. Derate linearly at 0.58 A dc/°C for T_C>125°C.

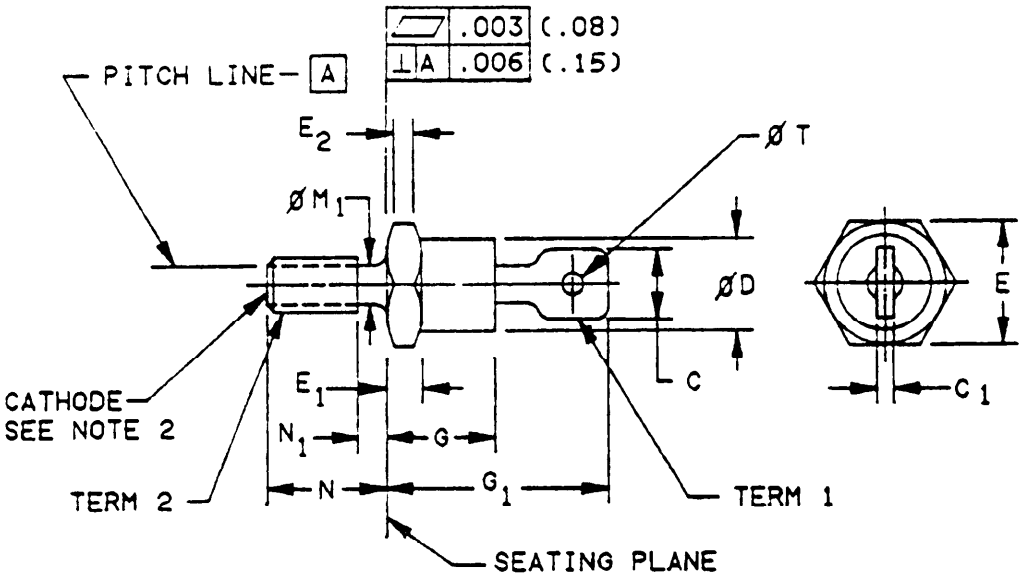
1.4 Primary electrical characteristics at T_C = 25°C, unless otherwise noted.

Type	Max V_{FM1} I_{FM} = 50 A (pk)	Max V_{FM2} I_{FM} = 5 A (pk)	Max I_{RM} V_{RM} = 45 V (pk) T_J = 175°C	Max I_{RM} V_{RM} = 45 V (pk) T_J = 25°C	Max C_T V_R = 5 V dc	Max $R_{\theta JC}$
	V (pk)	V (pk)	mA (pk)	mA (pk)	pF	°C/W
1N6391	0.68	0.44	400	15	2000	2.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5961

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
C		.250		6.35	1
C_1	.018	.065	.46	1.65	
ØD	.265	.424	6.74	10.76	
E	.424	.437	10.77	11.10	
E_1	.075	.175	1.91	4.44	
E_2	.060	.175	1.53	4.44	3
G	.300	.405	7.62	10.28	
G_1	.600	.800	15.24	20.32	
$ØM_1$	.163	.189	4.14	4.80	
N	.422	.453	10.72	11.50	
N_1		.078		1.98	4
ØT	.060	.103	1.52	2.62	

NOTES:
1. Angular orientation and contour of this terminal is undefined.
2. Pitch diameter 10-32 UNF-2A (coated) -.1697 (4.31 mm).
3. Chamfer or undercut on one or both ends of hexagonal portions is optional.
4. Length of incomplete or undercut threads of $ØM_1$.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
6. Metric equivalents are in parentheses.

FIGURE 1. Semiconductor device, diode type 1N6391, JAN, JANTX, and JANTXV.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standard, of the issued listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 Test Methods for Semiconductor Devices.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.2.1 Polarity. Devices shall have the cathode electrically connected to the stud.

3.3 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin may be omitted from the body of the rectifier.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV).1/ Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels
3	Test condition F; T_{low} = -55°C; T_{high} = +175°C
4	Not applicable
10	Not applicable
11	Reverse energy test (see 4.5.2) and table I, subgroup 2 herein.
12	MIL-STD-750, method 1038, test condition A, T_C = 150°C V_{RM} = 36 V (pk); I_O = 0; f = 60 Hz,; Alternate test: V_R = 36 V dc and T_C = 150°C
13	Subgroup 2 of table I herein: Δ V_{FM1} = ±0.05 V (pk) Δ I_{RM1} = ±100% or 5 mA dc whichever is greater Scope display evaulation (see 4.5.3)

1/ All JAN level devices shall be subjected to the scope display evaluation test.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb of MIL-S-19500 and table II herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Peak reverse energy test. The peak reverse energy test is to be performed as shown on figure 2. The Schottky rectifier under test must be capable of absorbing the reverse energy, as defined, and meet the electrical requirements of table I, subgroup 4 herein.

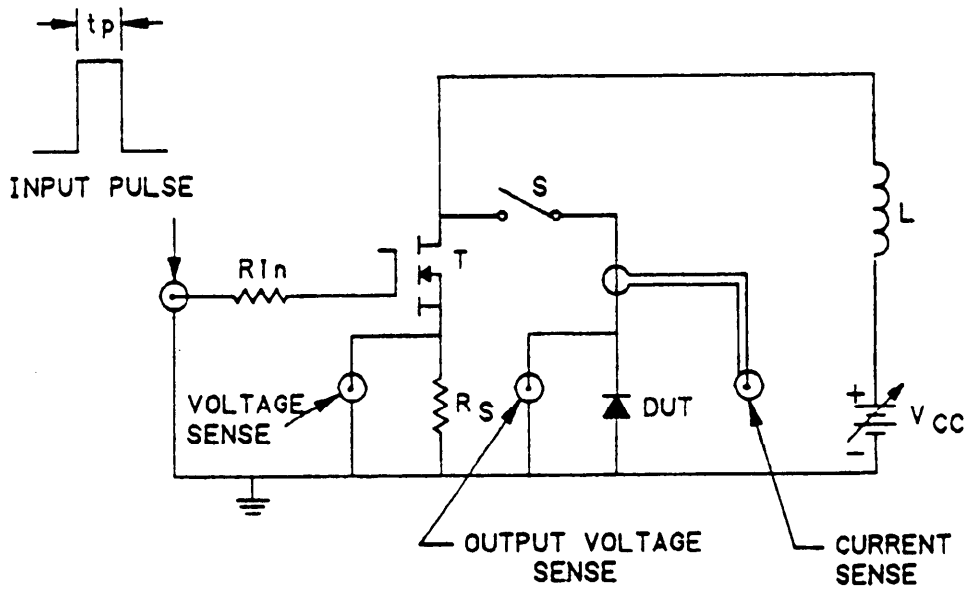
4.5.3 Scope display evaluation. A curve trace visual display of the reverse characteristics (voltage vs current) shall be made using a calibration of 10 mA per division vertical sensitivity, and 10 V per division horizontal sensitivity. Increase the reverse voltage until the reverse current reaches 50 mA minimum to 70 mA maximum. Observe the curve characteristics. Any device with an erratic pattern, such as double trace, other than capacitance effects, intermittance, discontinuities other than zener noise at the knee of the curve, or shorts shall be removed from the lot. Time limit for this test shall be 5 seconds minimum.

This test shall be the last test performed on the device in the 100% screening tests. The percent defective result for evaluation against the percent defective allowed (PDA) shall be determined prior to the results of this test.

4.5.4 Thermal resistance. Thermal resistance measurements shall be performed in accordance with MIL-STD-750, method 4081. The case reference temperature shall be held to equilbrium within the range of 20°C to 70°C during the power application, and shall be measured at the hex flat. The maximum limit of $R_{\theta JC}$ shall be 2.0°C/W. The following parameter measurements shall be used:

I_{F1} = 10 A dc minimum to 30 A dc maximum
I_{F2} = 100 mA dc

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Input Pulse
V_G = 10 Volts
R_G = 50 onms
P.W. $\approx$ 30 µs
Duty Cycle $\leq$ 1%

Rin = 50 ohms, 1 watt
R_S = 0.1 ohms, 1 watt
$V_{CC} \approx$ 10 volts
L = 260 µH
T = IRF130/2N6756 or equivalent.

Procedure: 1) With S open, adjust pulse width to test current of 2 amps across R_S.
2) Close S, Verify test current with current sense.
3) Read peak output voltage (see 4.5.5).

FIGURE 2. Peak reverse energy test circuit.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1			5				
Visual and mechanical examination	2071						
Subgroup 2			5				
Forward voltage	4011	I_{FM} = 50 A (pk) Pulsed (see 4.5.1)		V_{FM1}	---	0.68	V (pk)
		I_{FM} = 5 A (pk) Pulsed (see 4.5.1)		V_{FM2}	---	0.44	V (pk)
Reverse current leakage	4016	Pulse method (see 4.5.1); V_{RM} = 45 V (pk)		I_{RM1}	---	15	mA (pk)
Subgroup 3			5				
High temperature operation:							
Reverse current leakage	4016	Pulse method (see 4.5.1); V_{RM} = 45 V (pk)					
		T_J = 175°C		I_{RM2}	---	400	mA (pk)
		T_J = 125°C		I_{RM3}	---	40	mA (pk)
Low temperature operation:		T_C = -55°C					
Reverse current leakage	4016	Pulse method (see 4.5.1); V_{RM} = 45 V (pk)		I_{RM4}	---	400	mA (pk)
Forward voltage	4011	I_{FM} = 5 A (pk) Pulsed (see 4.5.1)		V_{FM3}	---	0.55	V (pk)
Subgroup 4			5				
Reverse current leakage at peak reverse voltage	4016	Pulse method (see 4.5.1); V_{RSM} = 54 V (pk) (alternate test see 4.5.5)		I_{RM5}	---	2.0	A (pk)
Capacitance	4001	V_R = 5 V dc; 100 KHz $\leq f \leq$ 1 MHz		C_T	---	2000	pF
Subgroups 5, 6, and 7							
Not applicable							

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TABLE II. Group B inspection for JAN, JANTX, and JANTXV.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			10
Thermal shock (temperature cycling)	1051	Test condition F with T_{low} = -55°C, T_{high} = 175°C	
Hermetic seal	1071		
Fine leak		Test condition G or H, leak rate $\leq 1 \times 10^{-7}$ atm cm^3/s	
Gross leak		Test condition A, C, D or F	
Electrical measurements		See table I, subgroup 2	
Subgroup 3			5
Steady-state operation life	1027	T_C = 150°C, V_{RM} = 36 V (pk) or V_R = 36 V dc	
		Alternate condition: T_C = 150°C I_O = 8 A dc, V_{RWM} = 45 V (pk)	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			
Decap, internal visual (design verification)	2075	Visual criteria in accordance with qualified design.	1 device /0 failure for each lot
Subgroup 5			
Not applicable			
Subgroup 6			7
High temperature life (nonoperating)	1032	T_A = 175°C	
Electrical measurements		See table I, subgroup 2	
Subgroup 7			10
Surge current	4066	I_{FSM} = 600 A (pk); V_{RM} = 45 V (pk) I_O = 22.5 A dc	
		10 surges of 8.3 ms each at 1 minute intervals	
		100°C < T_C < 125°C	
Electrical measurements		See table I, subgroup 2	

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TABLE III. Group C periodic inspection (all quality levels).

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Physical dimensions	2066	See figure 1	
Subgroup 2			10
Thermal shock (glass strain)	1056	Test condition B	
Terminal strength (tension)	2036	Test condition A; 10 lb; t = 15 s	
Terminal strength (bending stress)	2036	Test condition F (method B); t = 15 s	
Terminal strength (term. torque)	2036	Test condition D1; 10 oz-in; t = 15 s	
Terminal strength (stud torque)	2036	Test condition D2; 15 lb-in; t = 15 s	
Hermetic seal	1071		
Fine leak		Test condition G or H, leak rate $\leq 1 \times 10^{-7}$ atm cm^3/s	
Gross leak		Test condition A, C, D or F	
Moisture resistance	1021	Omit initial conditioning	
External visual	2071		
Electrical measurements		See table I, subgroup 2	
Subgroup 3			10
Shock	2016		
Vibration variable frequency	2056		
Constant acceleration	2006	10,000 G minimum	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			15
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			λ = 10
Steady state operation life	1026	I_O = 22.5 A dc, V_{RM} = 36 V (pk), T_C = 125°C ±3°C	
Electrical measurements		See table I, subgroup 2	
Subgroup 7			15
Operational power cycling (see 4.5.6)		I_O = 20 A dc; T_C(low) = 40°C + 0, -15°C T_C (high) = 115°C +5, -0°C 5000 cycles	
Electrical measurements		See table I, subgroup 2	
Subgroup 8			15
Thermal resistance	4081	See 4.5.4	

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4.5.5 Reverse current at peak reverse voltage, alternate test. The reverse current at peak reverse voltage test requirement may be satisfied by performing the reverse energy test of 4.5.2 and measuring breakdown voltage to ensure $V_{BR} \geq 54$ V (pk) with I_{RM} = 2.0 A (pk).

4.5.6 Operational power cycling. One complete cycle for this test shall consist of the following two steps: Step 1, heat the case to the T_C (high) specified by passing the specified forward current through the diode under test. The reverse voltage shall be only enough to permit the reverse current to flow, and in any event should be a maximum of 5 volts. Step 2, remove the applied current and allow the case temperature to cool to the T_C (low) specified. No time limit is applicable to the steps, but the cycling must be continuous until the required number of cycles has been completed. It is permissible to force cool the device during step 2.

This test shall be considered a qualification test only, and needs to be performed during initial qualification and as applicable to verify the adequacy of a design change.

4.5.7 Inspection conditions. Unless otherwise specified in MIL-S-19500 or herein, all inspections shall be conducted at a case temperature (T_C) of 25°C ±3°C.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Air Force - 13, 15

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-19500/553(USAF) Semiconductor Device, Diode, Silicon, Schottky Barrier

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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The documentation and process conversion measures necessary to comply with this amendment shall be completed by 9 June 1994

INCH-POUND

MIL-S-19500/553
AMENDMENT 6
9 March 1994
SUPERSEDING
AMENDMENT 5
3 November 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,
SCHOTTKY BARRIER, FAST RECOVERY,
TYPE 1N6391
JAN, JANTX, JANTXV, and JANHC

This amendment forms a part of MIL-S-19500/553, dated 28 May 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1, end of paragraph, add: "One level of product assurance for die (element evaluation).".
* 1.2, after "DO-4" Add: ", see figure 1A for JANHC die dimensions."
1.4, Max V_{FM1} column: Delete "0.72" as added in Amendment 2 and substitute "0.68".
1.4, Max V_{FM2} column: Delete "0.48" as added in Amendment 4 and substitute "0.50".

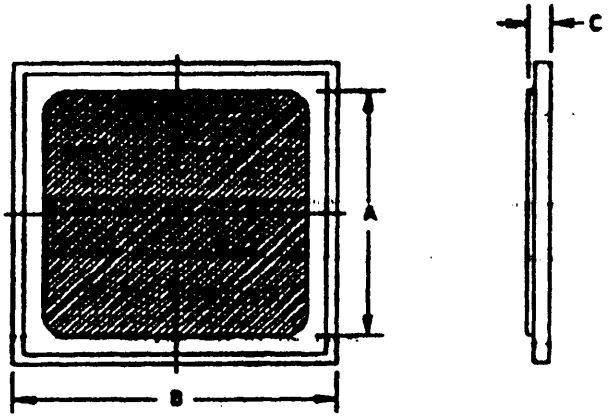
PAGE 2

* Add the following new figure 1A:

"

A Version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.167	.177	.53	.63
B	.175	.185	.33	.43
C	.012	.013	.305	.330



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are: Metallization (front (anode), consists of: W thickness = 3,600 Å minimum, Pt thickness = 1,800 Å minimum, Ag thickness = 30000 Å minimum. Metallization (back, (cathode)) consists of: Ti thickness = 1,600 Å minimum, NI thickness = 4,000 Å minimum, Ag thickness = 4000 Å minimum.

FIGURE 1A. JANHCA die dimensions."

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PAGE 3

* 3.2, line 2: Delete "figure 1" and substitute "figures 1 and 1A".

* Add the following new paragraphs:

"4.2.1 JANHC devices. Qualification for JANHC devices shall be in accordance with appendix H of MIL-S-19500."

"4.3.1 Screening (JANHC). Screening of JANC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed to ensure compliance with group A, subgroup 2."

PAGE 6

TABLE I, forward voltage, V_{FM1}, Max limits column: Delete "0.72" as added in amendment 2 and substitute "0.68".

TABLE I, forward voltage, V_{FM2}, Max limits column: Delete "0.48" as added in amendment 4 and substitute "0.50".

TABLE I, forward voltage, V_{FM3}, Max limits column: Delete "0.59" as added in amendment 4 and substitute "0.60".

PAGE 7

TABLE II , subgroup 2, hermetic seal, fine leak, conditions column: Delete and substitute "Test condition G or H".

PAGE 8

TABLE III, subgroup 2, terminal strength (bending stress), conditions column: Delete "t = 15 s" and substitute "5 pounds; t = 15 s".

TABLE III, subgroup 2, hermetic seal, fine leak, conditions column: Delete and substitute "Test condition G or H".

PAGE 9

* Add the following new paragraph:

"6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Case outlines (see figures 1 and 1A).

d. For die acquisition, the JANHC letter version shall be specified (see figure 1A).

e. Product assurance level and type designator."

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* Add the following new paragraph and table:

" 6.3 Suppliers of JANHC die. The qualified JANHC suppliers with the applicable letter version (example, JANHCA1N6391) will be identified on the QPL.

JANHC ordering information	
	Manufacturer
PIN	59377
1N6391	JANHCA1N6391

"

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

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NASA - NA

Review Activities:
Air Force - 13, 15, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-F119)

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